Back-End Home Assignment – Python Execution Agent

Task Overview

Build a minimal execution agent that can receive Python code, run it safely in an isolated environment, and return output or errors. The purpose is to evaluate architectural thinking, safety considerations, modularity, and API design. Expected duration: ~1 hour.

Requirements

1. API Endpoints

POST /execute – Accepts a JSON payload containing Python code:
{ "code": "print(2+2)" }

Returns stdout and stderr as JSON.

GET /health – Returns simple health status and timestamp.

2. Execution Agent Behavior

- Must execute Python code in an isolated manner.
- Must enforce execution timeout.
- Must capture stdout and stderr.
- Must handle Exceptions.

3. Architecture Expectations

The project must not be a single file. Expected structure:

```
/app
  /routes
  /services
  /models
  /utils
main.py
```

The execution logic must be separated from the API layer.

4. Technology

Backend framework: Python OR .NET Core.
Execution engine: Python code execution.

Evaluation Criteria (0–5 each)

1. Architecture & Separation of Concerns – Is the service well structured and scalable?
2. Safety & Isolation – Did the candidate avoid unsafe exec()? Are timeouts implemented?
3. Code Quality – Readability, naming, organization, maintainability.
4. Error Handling – Syntax errors, runtime errors, infinite loops, malformed JSON.
5. Bonus – Async execution, container abstraction, logging, typing, tests.

Deliverables

Candidate will present their solution in a 1-hour live meeting followed by Q&A. No code submission is required beforehand. The candidate should be prepared to explain:

- Design decisions
- Safety considerations
- How they would scale this into a production-ready execution engine